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Stored Functions/Functions:
Function is a block of code similar to a
stored procedure which is also used to perform an action
and returns result as a value.
Function can be divided into two types, these are
1) Scalar-Valued Function: In this case we can return a
attribute datatype as an output from the function.
Syntax:
Create Function <Function Name> (@parameter <Data Type> [size])
Returns <return attribute data type>
As Begin
<Function Body>
Return <return attribute name>
End

```

How to Call Scalar valued Functions:

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Syntax:      Select dbo.<Function Name> (value)
1) Create a function to return the cube of the given value.
create function fcube (@x int) returns int
as begin
return @x*@x*@x end

```

Output:select dbo.fcube(3)

2) Create a function that takes an employee id and returns the salary of that employee.

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create function fsal(@eid int) returns money
as begin
declare @sal money
select @sal=salary from employee where empid=@eid return @sal
end

```

Output:select dbo.fsal(1)

2) Table-Valued Function: In this case we can return a table as an output from the function.

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Syntax:
Create Function <Function Name> (@parameter <Data Type> [size]) Returns <Table>
As
Return <return select statement>

```

How to Call a Table-Valued Function:

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Syntax:      select * from functionname(value)
Ex: Create a function that accept the Address and returns the list of employee working
in given address from the table.
create function ft1(@add varchar(50)) returns table
as
return(select * from employee where address=@add)

```

Output:select \* from ft1('hyd')

Ex:Create a function to get the deptno and return list of employee working in EMP and DEPT tables.

```

create function saidata(@deptno int) returns table
as
return(select e.eid,e.ename,e.salary,d.deptno,d.dname,d.location from emp e inner join
dept d on e.deptno=d.deptno where e.deptno=@deptno)
Output: Select * from saidata(10)

```

How To Drop Functions:

- Drop Function <Function Name>

Ex: Drop Function Saidata

Difference between Function And Procedure:

- A function must return a value where as procedure never returns a value.
- A procedure can have parameters of both input

(with parameters) and output (without parameters)  
where as a function can have only input (with parameters)  
parameters only.

- In procedure we can perform select, insert, update and delete operation where as function can used only to perform select. Cannot be used to perform insert, update and delete operations.
- A procedure provides the option for to perform transaction management where as these operations are not permitted in a function.
- We call a procedure using execute command where as function are called by using select command only.

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